

QUANTIVOX AI FINANCIAL SENTIMENT

Market Sentiment Tracking & Comparative Sentiment Analysis for Stocks and Crypto Assets - Business Analytics Capstone Project Documentation Report

1. Introduction

Financial markets are heavily influenced by investor sentiment, financial news, economic developments, and geopolitical events. With the rapid growth of digital financial platforms, analyzing textual market information has become increasingly important. Traditional market analysis mainly focuses on numerical indicators such as stock prices and trading volumes, but these methods often fail to capture the emotional and psychological aspects of market behavior.

Natural Language Processing (NLP) and Artificial Intelligence (AI) have emerged as effective technologies for extracting sentiment from financial text data. Financial news headlines provide valuable insights into market optimism, fear, uncertainty, and investor expectations, helping analysts better understand market trends and behavior.

Quantivox AI Financial Sentiment is a Business Analytics and NLP-based financial intelligence platform developed to monitor and visualize market sentiment using financial news data. The system applies VADER and BERT sentiment analysis models to classify financial headlines into five sentiment categories ranging from Very Negative to Very Positive. The platform also integrates sector comparison, sentiment trend analysis, volatility monitoring, and interactive dashboard visualizations to support AI-driven financial market intelligence and business analytics. The project demonstrates the practical application of Business Analytics, Artificial Intelligence, Financial Analytics, and Data Visualization techniques in creating an interactive financial sentiment intelligence system.

2. Problem Statement

Modern financial markets generate massive volumes of news and information on a continuous basis. Investors and analysts face considerable difficulty in manually processing large-scale financial content and identifying meaningful sentiment trends in real time. Market sentiment changes rapidly due to earnings reports, economic announcements, industry developments, and global events, making it challenging to monitor market psychology effectively.

Existing analytical systems primarily focus on historical price analysis and quantitative indicators, while providing limited support for sentiment-driven analytics. Many platforms lack advanced NLP integration, sector-wise comparison, multi-ticker sentiment monitoring, and unified stock and cryptocurrency analysis.

As a result, there is a growing need for an AI-powered financial sentiment intelligence platform capable of:

- Automatically collecting financial news data
- Performing NLP-based sentiment classification
- Monitoring overall market sentiment trends
- Comparing sentiment across industries and tickers
- Providing interactive visual analytics for investment research and decision-making

Quantivox AI Financial Sentiment addresses these limitations by integrating NLP models, financial analytics, and interactive dashboard visualization into a single business intelligence platform.

3. Project Objectives

The primary objective of this project is to develop an NLP-driven Financial Sentiment Intelligence System capable of analyzing financial news headlines and transforming textual information into meaningful market insights.

The project focuses on sentiment trend analysis, ticker-wise comparison, industry-level monitoring, and overall market intelligence using Artificial Intelligence and Business Analytics techniques.

The major objectives of the project include:

- Developing an AI-powered financial sentiment analysis platform
- Performing sentiment classification using VADER and BERT models
- Monitoring market sentiment trends across multiple industries
- Enabling comparative analysis between stocks and cryptocurrencies
- Visualizing financial sentiment using interactive dashboards
- Identifying bullish and bearish market conditions
- Supporting business analytics and investment research activities
- Generating exportable analytical reports and datasets

The project also aims to demonstrate the integration of NLP, data analytics, and visualization technologies in solving real-world financial analysis problems.

4. Scope of the Project

The scope of Quantivox AI Financial Sentiment is centered on financial news sentiment analysis and market intelligence visualization. The project focuses on extracting, processing, analyzing, and visualizing financial textual data to monitor investor sentiment and market behavior.

The platform supports both stock market and cryptocurrency sentiment analysis and provides comparative insights across industries, market sessions, and ticker categories.

The project scope includes:

- Financial news collection from Finviz
- Stock and cryptocurrency sentiment analysis
- NLP-based sentiment classification
- Ticker-wise and industry-wise sentiment tracking

- Sentiment volatility monitoring
- Comparative VADER and BERT analysis
- Interactive dashboard visualization
- Export of processed analytical data

The project does not include predictive stock forecasting, automated trading execution, portfolio optimization, or real-time social media analytics. The primary emphasis remains on sentiment intelligence and business analytics visualization.

5. Dataset & Data Sources

The project uses financial news headlines and market data collected from reliable online financial platforms.

Financial news headlines are collected from the Finviz platform using web scraping techniques implemented through BeautifulSoup. Finviz provides structured financial news related to stocks, industries, and cryptocurrencies, making it suitable for large-scale sentiment analysis.

Market-related ticker and price information is collected using the yFinance Python library. This enables the system to associate financial news with corresponding stock and cryptocurrency assets.

The project focuses on five major industry categories:

1. Technology – Semiconductor & AI
2. Financial Services – Banking & FinTech
3. Energy – Oil & Gas
4. Healthcare – Pharma & Biotechnology
5. Consumer & E-Commerce

A diversified set of companies from each sector is included to ensure balanced sentiment analysis and industry-level comparison.

6. Selected Companies & Industry Coverage

The system includes major publicly traded companies representing different sectors of the financial market.

The Technology sector includes companies such as NVIDIA, AMD, Intel, Taiwan Semiconductor Manufacturing Company, Broadcom, Qualcomm, Micron Technology, ASML, Arm Holdings, and Applied Materials.

The Financial Services sector includes JPMorgan Chase, Bank of America, Goldman Sachs, Morgan Stanley, Citigroup, SoFi Technologies, PayPal, Visa, Mastercard, and Nu Holdings.

The Energy sector includes Exxon Mobil, Chevron, ConocoPhillips, Schlumberger, Halliburton, BP, Shell, Occidental Petroleum, EOG Resources, and Baker Hughes.

The Healthcare sector includes Pfizer, Johnson & Johnson, Merck & Co., AbbVie, Eli Lilly and Company, Bristol Myers Squibb, Gilead Sciences, Moderna, Regeneron Pharmaceuticals, and Amgen.

The Consumer & E-Commerce sector includes Amazon, Walmart, Costco Wholesale, Target Corporation, Alibaba Group, MercadoLibre, Etsy, Shopify, eBay, and JD.com.

The inclusion of multiple sectors allows the dashboard to perform comparative sentiment analysis across industries and identify sector-specific market behavior.

7. Analytics & NLP Approach

The project applies Business Analytics and Natural Language Processing (NLP) techniques to transform unstructured financial news into actionable market intelligence.

The analytics framework focuses on:

- Descriptive analytics to summarize overall market sentiment
- Comparative analytics to compare industries, tickers, market sessions, and NLP model outputs
- Visual analytics using dashboards, charts, heatmaps, histograms, and trend analysis for business interpretation

Key business metrics include: Average Sentiment, Market Volatility, News Volume, Bullish/Bearish Trends, Sector-wise Sentiment Comparison

The project integrates two NLP models:

VADER : A rule-based sentiment analysis model optimized for short financial headlines and fast polarity detection.

BERT / FinBERT : A transformer-based deep learning model that understands contextual financial language and provides more accurate sentiment classification.

Both models classify sentiment into: Very Positive, Positive, Neutral, Negative, Very Negative

The combined VADER and BERT approach enables comparative sentiment analysis and improves the reliability of financial sentiment intelligence.

9. Data Preprocessing & Feature Engineering

Data preprocessing is performed to improve the quality, consistency, and usability of financial news data before sentiment analysis. The preprocessing stage includes removal of duplicate records, null value handling, textual normalization, timestamp conversion, and headline cleaning.

Duplicate records are removed using the combination of ticker and headline fields to avoid repeated sentiment analysis on identical news entries. Feature engineering is then applied to create additional analytical variables that support business intelligence visualization.

The engineered features include:

- Ticker symbol
- Industry classification
- Date and hour of publication
- Financial quarter
- Market session category
- Cryptocurrency identifier (`is_crypto`)
- VADER sentiment score
- BERT sentiment score
- VADER sentiment class
- BERT five-level sentiment class

These engineered features enable advanced filtering, trend analysis, sector comparison, and dashboard interactivity.

10. Dashboard Design & Visualization

The Quantivox dashboard is designed using a dark-themed professional financial analytics interface focused on readability, clarity, and analytical interpretation.

The dashboard provides a centralized view of market sentiment using KPI cards, trend charts, heatmaps, histograms, scatter plots, and sector comparison visuals.

The design follows a sentiment-based color framework where:

- Green represents bullish or positive sentiment
- Yellow represents neutral sentiment
- Red represents bearish or negative sentiment
- Orange represents moderately bearish conditions
- Light green represents moderately bullish conditions

The major dashboard components include:

- Quantivox Sentiment Index (QSI)
- Market Sentiment Gauge
- Average Sentiment KPI
- Momentum KPI
- Volatility KPI
- News Volume KPI
- Top Bullish Stock Indicator
- Most Impact Stock Indicator
- Daily Sentiment Trend Line
- Sector Comparison Chart
- Sentiment Distribution Histogram
- VADER vs BERT Scatter Plot
- Market Session Analysis
- Hourly News Activity Chart
- Word Cloud Visualization
- Sector Sentiment Heatmap

The dashboard is developed to support interactive business analytics and real-time market sentiment interpretation.

Dashboard Analytics Summary :

Section	Component	Details
Market Index (QSI)	Formula	$QSI = ((Average\ Sentiment + 1) / 2) * 100$
Market Index (QSI)	Interpretation Scale	0–20 Bearish, 21–40 Moderately Bearish, 41–60 Neutral, 61–80 Moderately Bullish, 81–100 Bullish
Market Visualization	Sentiment Gauge	Shows overall market sentiment from Fear → Greed
Market Visualization	Daily Trend	Tracks sentiment movement over time
Market Visualization	Sector Analysis	Compares sentiment across industries
Market Visualization	Industry Distribution	Shows news share across sectors
Sentiment Analytics	Distribution Analysis	Histogram of VADER & BERT sentiment scores
Sentiment Analytics	Model Comparison	Scatter plot showing VADER vs BERT agreement
Sentiment Analytics	Market Session Analysis	Sentiment across trading sessions
Sentiment Analytics	News Activity Tracking	Hourly news publication volume
Text Analytics	Word Cloud	Most frequent financial keywords
Data Exploration	Dataset View	Filtered raw dataset display
Statistical Testing	Objective	Correlation between VADER and BERT
Statistical Testing	Formula	$corr = df['vader_score'].corr(df['bert_score'])$
Statistical Testing	Hypothesis (H ₀)	No relationship between models

Statistical Testing	Hypothesis (H ₁)	Significant relationship exists
Statistical Testing	Output	Accept/reject hypothesis with conclusion
KPI Engine	Average Sentiment	df['vader_score'].mean()
KPI Engine	News Volume	len(df)
KPI Engine	Stock Sentiment Ranking	groupby mean idxmax/idxmin
KPI Engine	Trend Analysis	groupby date mean
KPI Engine	Activity Analysis	value_counts()
KPI Engine	Volatility	std()